Hall Ticket Number:

MR-21

Code: B414CB4S2

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, July/Aug-2024

Object Oriented Programming Using Java

(Common to CSB, CSM & CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

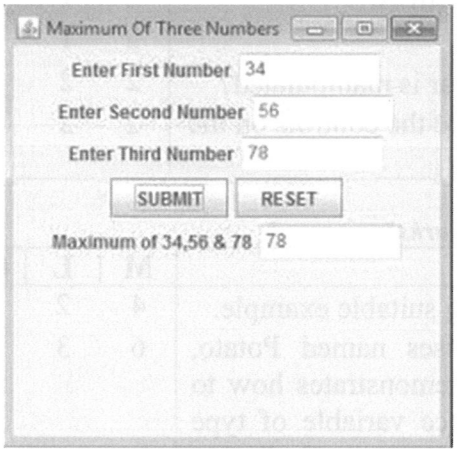
Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	List the data types available in Java.	2	1	1	1,2
1. b)	Can a method be declared as both abstract and final? Justify your answer.	2	1	1	1,2
1. c)	Write about the variables in the interface.	2	1	2	1,2
1. d)	How are exceptions handled in Java?	2	1	2	1,2
1. e)	Define Multithreading.	2	1	4	1,2
1. f)	Differentiate between Byte streams and character streams. Give any two examples for each stream.	2	1	3	1,2
1. g)	In which situation ArrayList is preferred over LinkedList?	2	1	3	1,2
1. h)	List the legacy classes available in Java.	2	1	3	1,2
1. i)	Mention the event object created when the scroll bar is manipulated?	2	2	5	1,2
1. j)	Is it mandatory to use the Layout Managers to place the controls on the window? Explain your answer.	2	2	5	1,2

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Define class. Explain constructor overloading using suitable example.	4	2	1	1,2,3
2. b)	The abstract class Vegetable has three subclasses named Potato, Brinjal and Tomato. Write a java program that demonstrates how to establish this class hierarchy. Declare one instance variable of type String that indicates the color of a vegetable. Create and display instances of these objects. Override the toString() method of Object to return a string with the name of vegetable and its color.	6	3	1	1,2,3
	OR				
2. c)	Differentiate between Compile time and Runtime Polymorphism.	4	2	1	1,2
2. d)	Define Time class with hour and minute as data members. Also define display method to display the time and addition method to add two time objects. Write the main () method to test the above class methods.	6	3	1	1,2,3
3. a)	What is a package? Mention the different access protection using packages.	6	2	2	1,2
3. b)	What is Exception? Demonstrate how you can handle different types of exception separately.	4	2	2	1,2
	OR				
3. c)	Explain the difference between an abstract class and an interface using appropriate examples.	4	2	2	1,2,3
3. d)	Write a Java program to illustrate user-defined Exception.	6	2	2	1,2,3
4. a)	Write a program to copy the contents of one file into another file.	5	3	3	1,2,3
4. b)	What is Multithreading? What are the different ways to create Threads? Write a program to demonstrate multithreading.	5	2	4	1,2,3
	OR				

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4. c)	Write a program to create 2 threads and set the priorities of the threads.	5	3	4	1,2,3
4. d)	Write a program to create a file that can store details about the employees of an organization and display the details of employees who belong to CSE department. Details include empId, empName, empAge, empDept, and empSal. Read the data through keyboard. Use serialization and De-serialization.	5	3	3	1,2,3
5. a)	Write a program to store the student records in a LinkedList and display the second record.	4	3	3	1,2,3
5. b)	What is the need of StringTokenizer class? Write a program to display the words in the given sentence.	6	3	3	1,2,3
	OR				
5. c)	Explain the Map interfaces and classes.	6	2	3	1,2
5. d)	ListIterator and Iterator are the two ways to iterate through a Collection. Is there another way? Explain with example.	4	2	3	1,2,3
6. a)	Explain MVC Architecture.	2	2	5	1,2
6. b)	Create an Applet, with the UI as shown below. Implement Event handling on each of these buttons, one to find the maximum of the given three no's in the textfields and the other to reset the values to 0. The maximum value has to be displayed in the textfield below the buttons.  	8	3	5	1,2,3
	OR				
6. c)	List the different Layout managers available in java. Explain any 3 layout managers with examples.	7	2	5	1,2
6. d)	Differentiate between Swing and AWT.	3	4	5	1,2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome